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Automatic Assessment of Speech Production Skills for Children with Cochlear Implants Using Wav2Vec2.0 Acoustic Embeddings

Seonwoo Lee[1], *Sunhee Kim*[1], *Minhwa Chung*[1]

[1]Seoul National University, Republic of Korea
{lsw5220,sunhkim,mchung}@snu.ac.kr

Abstract

This study introduces an automatic assessment model for speech production skills of children with cochlear implants (CIs) to support home-based speech therapy. The model employs acoustic embeddings from self-supervised models and considers speech traits of both normal hearing (NH) adults and children, which is a novel method for evaluating speech of children with disorders. It combines phoneme embeddings and two acoustic embeddings from Wav2Vec2.0 models, each trained on the speech of NH adults and children, via multi-head attention. Using a speech corpus of Korean-speaking children with CIs, our model outperforms single-embedding methods in a Pearson correlation coefficient between predicted and expert-rated scores, with a relative improvement of 51%. The results highlight the effectiveness of Wav2Vec2.0 acoustic embeddings and the importance of incorporating both of typical speech patterns of NH adults and children in assessing speech production skills in children with CIs.

Index Terms: automatic assessment, speech production skills, children with cochlear implants, self-supervised models

1. Introduction

Cochlear implants (CIs) is a kind of medical intervention which can restore partial or functional hearing to individuals with hearing loss by stimulating the residual auditory nerve with electric nodes [1]. Despite improved speech perception, children with CIs often encounter issues related to speech production due to distorted speech perception and fixed incorrect articulation [2]. Their speech shows differences from those of children with normal hearing (NH) in the production of segmental parts of speech, including vowels and consonants, and in phonological process errors [3, 4]. Recent research on Korean-speaking children with CIs has shown that although speech intelligibility displayed no significant difference, the speech acceptability of children with CIs was lower than that of children with NH, with acoustic differences observed in pitch, voice quality, accentedness, resonance, and speech rate [5]. Less elaborate and stable movements of articulators [6] are also a distinct characteristic of these children.

To improve their speech production skills, (re)habilitation should include continuous training which extends from therapy sessions to the home environment where they spend most of their time. However, parents or guardians frequently face difficulties with home-based training due to limited knowledge in identifying errors in speech. For example, distinguishing which phonological errors need correction is challenging, since children with CIs exhibit developmentally atypical ones compared to children with NH as well as developmentally typical ones which are different from the speech of adults [3, 4]. The lack of expertise adds to their physical and mental burden. An automatic speech production assessment system could relive these challenges associated with home training.

In contrast to relatively active research on automatic assessment systems for speech of NH children in the fields such as reading ability assessment [7, 8] and foreign language learning [9], research on speech production assessment for children with disorders, especially for children with CIs, has been limited. For children with speech sound disorders, decoding graphs were extended to include mispronunciations produced by the target children [10]. For children with hearing loss, an automatic speech production assessment system utilizing neural network output activities was proposed to identify segmental errors [11]. Although these studies demonstrated the effectiveness of their methods in assessing speech production skills in children, there still remains some limitations. Primarily, only the segmental level of speech production was assessed [10, 11]. However, diverse aspects of speech characteristics such as articulation and prosody should be reflected to assess speech production skills of children with CIs [5]. Furthermore, direct consideration of speech differences between NH adults (hereafter referred to as adults) and children with NH was absent [9, 11].

Recently, pre-trained self-supervised models have been widely used for automatic pronunciation assessment in foreign language learning. These studies have employed acoustic embeddings extracted from fine-tuned models whose training objectives include automatic speech recognition [9, 12] or phoneme recognition [13, 14, 15]. These embeddings are effective because they encode features related to phonetic elements, such as consonants and vowels [16], as well as various prosodic features, including fluency, stress, and duration [17]. Specifically, [13] employed multi-head attention to combine language-specific embeddings, which serve as the assessment criterion with acoustic embeddings.

In this paper, we propose an automatic speech assessment model for Korean-speaking children with CIs, leveraging Wav2Vec2.0 acoustic embeddings. The model incorporates two acoustic embeddings, each extracted from acoustic models fine-tuned with adults' speech and NH children's speech. Adopting language-specific embeddings representing standard speech patterns of the native language in the foreign language learning area, these acoustic embeddings are expected to reflect standard speech characteristics. In addition, the differences between typical speech production characteristics of adults and children with NH are directly considered through attention mechanisms. The rest of the paper is organized as follows: the next section describes the proposed method, followed by an explanation of the experiments in Section 3. Section 4 reports experimental results, which is analyzed and discussed in Section 5. Section 6 presents the conclusion and future work.

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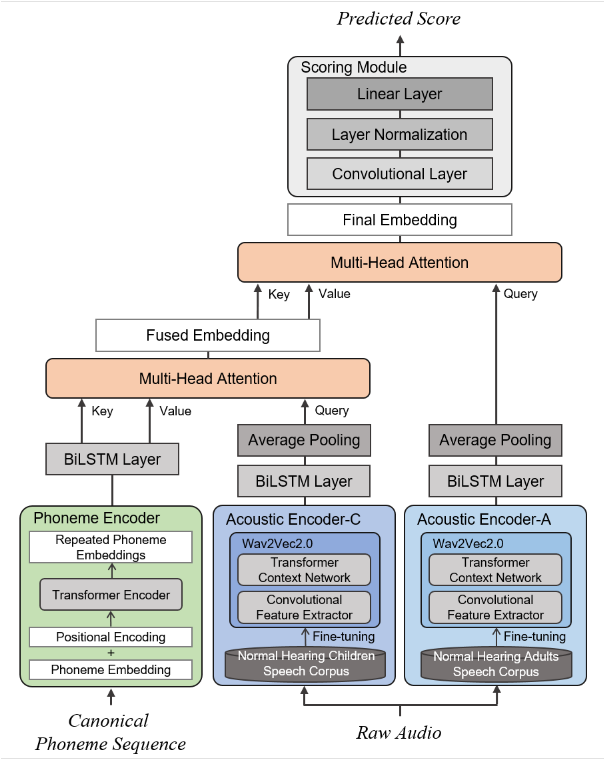

Figure 1: *The overview of the proposed method*

2. Method

The overview of the proposed method is shown in Figure 1. The model comprises three encoders, two multi-head attentions (MHA), and a scoring module. The encoders include a phoneme encoder, an acoustic encoder fine-tuned with adults' speech (Acoustic Encoder-Adults; AE-A), and an acoustic encoder fine-tuned with speech from children with NH (Acoustic Encoder-Children with NH; AE-C).

Canonical phoneme sequence and raw speech audio are input into the phoneme encoder and the acoustic encoders, respectively. The phoneme embeddings are initially processed through an MHA with the acoustic embeddings from AE-C. The resulting fused embeddings are then processed through another MHA with the acoustic embeddings from AE-A. The final embeddings are passed to the scoring module, where the predicted score is output through a convolutional layer, layer normalization, and a linear layer. The mean squared error loss function is used for training.

2.1. Phoneme Encoder

The phoneme encoder takes canonical phoneme sequences as input and outputs corresponding vectors. The phoneme embeddings are concatenated with positional embeddings, then fed into a transformer encoder to get contextualized embeddings. Since the length of the phoneme sequence is shorter than the number of speech frames, a process of repeating tensors in the contextualized embeddings is implemented to align with the dimension of acoustic embeddings. This repetition is determined by a factor calculated by dividing the embedding dimension by the phoneme sequence length.

2.2. Acoustic Encoders

Both acoustic encoders, AE-A and AE-C, are based on the multi-lingual Wav2Vec2.0 model, XLS-R-300m [18]. XLS-R is a large-scale model for cross-lingual speech representation learning, pre-trained with more than 436k hours of audio data from 128 languages. XLS-R-300m model contains over 300 millions parameters. Its architecture consists of a convolutional feature extractor and a transformer context network. Both encoders are fine-tuned for phoneme recognition using the Fairseq toolkit [19]. The last hidden states of the transformer block are extracted, then passed through a BiLSTM layer and an average pooling layer for multi-head attention.

AE-C encodes input speech from the perspective of the speech characteristics of children with NH. As children with NH exhibits developmental errors or different speech production compared to adults, the resulting encoded acoustic embeddings focus on developmentally atypical speech production of children with CIs, disregarding discrepancies related to developmental changes in children. Meanwhile, AE-A encodes input speech from the perspective standard adult speech characteristics, representing the desired form of speech production. Thus, the output embeddings contain information on developmentally typical speech patterns of children with NH and developmentally atypical speech patterns of children with CIs.

2.3. Multi-Head Attention (MHA)

Attention is a mechanism that maps a query and a set of key-value pairs to an output, focusing on the similarity or compatibility [20]. MHA is employed instead of a single attention function to jointly attend to features from different representation subspaces at different positions, capturing various relationships between the query and the key-value pairs. In the proposed model, a query generally represents the embeddings to be processed, primarily acoustic embeddings. A key-value pair is phoneme or acoustic embeddings to be attended to and selected in relation to the query. As MHA focuses on the segments of a key and a value which are most compatible with the query, the output embeddings contain more abundant information about the query.

The first MHA to form fused embeddings is applied between the phoneme embeddings as a key-value pair and the acoustic embeddings from AE-C as a query. The fused embeddings encode NH children's typical speech production matched with canonical phoneme sequences, focusing only on atypical patterns and disregarding developmentally typical ones. Then, another MHA is applied with the acoustic embeddings from AE-A as a query and the fused embedding as a key and a value to form the final embeddings. As the acoustic embeddings from AE-A contain information on both developmentally typical and atypical speech patterns, the final embeddings represent acoustic information regarding typical and atypical speech errors, with atypical ones additionally marked.

3. Experiments

3.1. Dataset

3.1.1. Speech Corpus of Korean-Speaking Children with CIs

A speech corpus for Korean children with CIs is utilized, which is being constructed, as there is no available speech corpus of children with CIs yet. All speakers in the corpus had their first implantation before the age of three. The audio data of speech from 30 children with CIs were extracted from videos taken during regular evaluation sessions for speech perception, speech production, and language skills at Seoul National University Hospital in Seoul, Korea. The extracted audio was segmented for individual utterances spoken by children with CIs.

Each utterance was orthographically transcribed for the target utterance. If it was different from the target prompt during speech perception evaluation, an additional orthographic transcription of the actual utterance was conducted. The actual utterances were then phonemically transcribed and rated on speech production skills by three nationally certified speech and language pathologists. Each speech and language pathologist was qualified with at least two years of experience in the rehabilitation of children with CIs or with knowledge of phonetics. Evaluation on speech production skills was conducted from the perspective of general speech production, considering the general aspects of speech including articulation, prosody, and voice quality. The score ranged from 1 to 5, with 1 indicating very poor, and 5 indicating very good speech production skills. All processes of constructing the corpus strictly adhered to ethical approval.

For experiments, speech from children with three years of experience with CIs is utilized. From the initial count of 2,279 (48 minutes(m)), any utterances longer than 2 seconds are excluded to reduce length variation, resulting in a total of 1,915 utterances (29m) from 14 children with CIs. The averaged speech production scores are used as the ground truth scores. The interrater reliability of the score assessed by the Intraclass Correlation Coefficient using SPSS version 26 is 0.733 with a 95% confidence interval of 0.711 to 0.753 ($p<0.001$).

3.1.2. Speech Corpora for Training Acoustic Encoders

The free conversation speech corpus of Korean-speaking NH children [21], a collection of spontaneous speech recordings of children whose age ranges from 3 to 10, is employed to fine-tune AE-C. For young children with difficulties in free conversation, repeating speech is also included. The duration of the audio amounts to more than 3,000 hours with more than 1,000 speakers. AE-C is fine-tuned with speech of NH children aged 3 to 5 in the free conversation speech corpus of Korean-speaking children because speech data employed for the experiments are from children with CIs who have used CI devices for three years. The training, validation, test dataset included 256,548 (240hours(h) & 16m), 12,121 (13h & 1m), 12,000 (10h & 54m) utterances, respectively.

KsponSpeech [22] is the dataset for fine-tuning AE-A, which consists of Korean spontaneous speech recordings from 2,000 native Korean speakers. This corpus contains 1,000 utterances per speaker during open-domain dialog of two speakers. Individual utterances are manually transcribed. The total duration amounts to 969 hours. For training AE-A, the training, validation, test dataset included 620,000 (965h & 9m), 2,545(3h & 56m), 3,000 (2h & 38m) utterances, respectively.

3.2. Experiment Details

The experiments are conducted via a 5-fold cross-validation approach, as the dataset of speech from children with CIs is quite small. Each fold is randomly split into 1,532 and 383 utterances for train and test sets, respectively. A speaker-dependent split is employed due to the varying number of utterances per child, which prevents imbalances in train/test data size and score distribution across folds.

For the phoneme encoder, the vocab size is 47 to include every phoneme in Korean and an extra one for space. The transformer encoder follows the default setting of the transformer architecture. The fine-tuning of acoustic encoders follows the default setting of XLS-R. The best phoneme error rate (PER) of the fine-tuned AE-C and AE-A on the test dataset achieved 8.80%, and 8.22%, respectively. The number of attention heads in MHA is set to 8. The embedding dimension for the entire model is set to 128. Adam optimizer is utilized with the learning rate 5e-4. Each model is trained during 50 epochs. Training a model with a fold using a GPU (NVIDIA GeForce RTX 2080 Ti) takes approximately 0.5 hours, resulting in 2.5 hours to complete training a model.

The baseline models are models utilizing single acoustic embeddings since there has been no study on automatic assessment of entire speech aspects of children with CIs. The experiments involve models that incorporate various combinations of three types of embeddings: phoneme embeddings, two acoustic embeddings from AE-C and AE-A, including the fusion of two types of embeddings and the sequential fusion of all three types. As the evaluation metric, Pearson correlation coefficients (PCC) are calculated between the averaged expert-labeled scores and predicted scores.

4. Results

4.1. Experimental Results

The experimental results are presented in Table 1 (above the solid line). The single-acoustic embedding approaches **A** and **C** display an average PCC of 0.4043 and 0.4983, respectively. The fusion of acoustic embeddings **A**$\otimes$**C** and **C**$\otimes$**A** shows an average PCC of 0.4228 and 0.4785, respectively. Attending each acoustic embeddings with phoneme embeddings leads to a higher PCC, 0.5599 for **P**$\otimes$**C** and 0.5645 for **P**$\otimes$**A**. The proposed method **(P**$\otimes$**C)**$\otimes$**A** achieves the best average PCC of 0.6116, with a relative improvement of 51.3% and 22.7% compared to **A** and **C**, respectively. Its standard deviation (SD) across the folds is 0.0149, showing higher stability. The second highest PCC is obtained by **P**$\otimes$**(C**$\otimes$**A)**, with an average PCC of 0.6075 and an SD of 0.147. The opposite configuration **P**$\otimes$**(A**$\otimes$**C)** also displays relatively high performance compared to the others.

4.2. Various Combinations of Embeddings

Since fusion of phoneme embeddings generally results in great improvement, various combinations of the embeddings fused with phoneme embeddings are explored as shown in Table 1 (below the solid line). Arithmetic operations involving **(P**$\otimes$**A)** and **(P**$\otimes$**C)** enhance the PCC. Further improvement is achieved by applying MHA.

Additional fusion is conducted to identify if using more complex representations would result in improvement, for example, to **(P**$\otimes$**A)**+**(P**$\otimes$**C)** (below the dashed line in Table 1). While most configurations show an increase in the average PCC, fusion with acoustic embeddings as a key and value decreases the performance. The configuration **(P**$\otimes$**C)**$\otimes$**[(P**$\otimes$**A)+(P**$\otimes$**C)]** yields a performance comparable to the best performance of the proposed model, with an average PCC of 0.6095.

5. Discussion

The configurations of single acoustic embeddings show a moderate correlation with an average PCC of more than 0.4, suggesting the effectiveness of Wav2Vec2.0 acoustic embeddings

Table 1: *Pearson correlation coefficients for predicted versus expert-labeled scores by embedding type.* ***A*** *refers to acoustic embeddings from the acoustic encoder-adults,* ***C*** *to acoustic embeddings from the acoustic encoder-children with normal hearing, and* ***P*** *to phoneme embeddings.* $\otimes$ *denotes multi-head attention, where the former represents a set of key and value pairs, and the latter represents a query.* $+$ *and* $-$ *denotes addition and subtraction of tensors, respectively. In all cases, p-values* < 0.001*.*

Embeddings	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Mean	SD
C	0.4578	0.4760	0.5310	0.5176	0.5091	0.4983	0.0304
A	0.3666	0.4314	0.3772	0.4573	0.3892	0.4043	0.0344
A⊗C	0.3842	0.3810	0.4210	0.4811	0.4467	0.4228	0.0425
C⊗A	0.4728	0.4329	0.4928	0.4961	0.4979	0.4785	0.0245
P⊗C	0.5595	0.6047	0.5836	0.5021	0.5495	0.5599	0.0388
P⊗A	0.5670	0.5800	0.5517	0.5440	0.5800	0.5645	**0.0146**
P⊗(A⊗C)	0.5757	0.5690	0.5558	0.5633	0.5979	0.5723	0.0161
P⊗(C⊗A)	0.5883	0.6089	0.5972	**0.6202**	**0.6227**	0.6075	0.0147
(A⊗C)⊗P	0.4054	0.3763	0.3971	0.4796	0.3993	0.4115	0.0396
(C⊗A)⊗P	0.4648	0.4819	0.5289	0.5135	0.5108	0.5000	0.0260
A⊗(P⊗C)	0.4666	0.4191	0.3972	0.4977	0.4790	0.4519	0.0422
C⊗(P⊗A)	0.4683	0.4492	0.4848	0.5324	0.4948	0.4859	0.0312
(P⊗A)⊗C	**0.6047**	0.5632	0.5282	0.5527	0.5630	0.5624	0.0276
(P⊗C)⊗A	0.5969	**0.6105**	**0.6347**	0.6005	0.6155	**0.6116**	0.0149
(P⊗A)+(P⊗C)	0.5584	0.6062	0.6033	0.5385	0.5708	0.5754	0.0335
(P⊗A)−(P⊗C)	0.5570	**0.6297**	0.5903	0.5319	0.5896	0.5797	0.0424
(P⊗C)−(P⊗A)	0.5673	0.6149	0.5937	0.5188	0.5759	0.5741	0.0414
(P⊗C)⊗(P⊗A)	0.5765	0.5787	**0.6284**	0.6077	0.6157	0.6014	0.0230
(P⊗A)⊗(P⊗C)	**0.6253**	0.5748	0.5593	0.5819	0.5994	0.5881	0.0253
[(P⊗A)+(P⊗C)]⊗A	0.5745	0.6006	0.5571	0.5761	0.6200	0.5857	0.0179
[(P⊗A)+(P⊗C)]⊗C	0.6008	0.5872	0.5491	0.6011	0.6159	0.5908	0.0245
[(P⊗A)+(P⊗C)]⊗(P⊗A)	0.5715	0.5957	0.5629	0.5646	0.6159	0.5821	0.0151
[(P⊗A)+(P⊗C)]⊗(P⊗C)	0.5611	0.5641	0.5742	0.5746	0.6092	0.5766	0.0069
A⊗[(P⊗A)+(P⊗C)]	0.4129	0.4070	0.4004	0.4680	0.4562	0.4289	0.0310
C⊗[(P⊗A)+(P⊗C)]	0.4258	0.4042	0.4120	0.3957	0.4513	0.4178	**0.0128**
(P⊗A)⊗[(P⊗A)+(P⊗C)]	0.6008	0.5887	0.5310	0.5811	0.5691	0.5741	0.0307
(P⊗C)⊗[(P⊗A)+(P⊗C)]	0.5877	0.5965	0.6264	**0.6184**	**0.6187**	**0.6095**	0.0182

in assessing speech production skills of children with CIs, as in previous studies on language learning [9, 12, 13, 14, 15]. Considering that the models for language learning have employed acoustic embeddings reflecting standard speech patterns in the native language, the better performance of **C** compared to **A** implies a more significant role of information related to atypical speech patterns in **C**. Meanwhile, the proposed method **(P⊗C)⊗A** obtains the best performance. This indicates that using acoustic embeddings that reflect the speech characteristics of either adults or NH children alone is insufficient, emphasizing the significance of leveraging both of them.

The fusion of phoneme embeddings generally increases the average PCC. As acoustic embeddings from self-supervised models represent features in comprehensive aspects of speech [16, 17], the fused embeddings would encode features beyond the frame-level processing of the acoustic model, capturing segmental and suprasegmental characteristics related to higher-level units, at least at the phoneme level. The improvement in performance of the second-best **P⊗(C⊗A)** can be attributed to the exclusively encoded information about both developmentally typical and atypical speech patterns, by incorporating phonemic information into the acoustic embeddings which consider developmental speech patterns. The relatively lower performance of the opposite configuration **P⊗(A⊗C)** could be due to developmental variability in NH children.

For queries, **A** seems more suitable than **C**, given that configurations involving **(C⊗A)** or **(P⊗C)⊗(P⊗A)** always perform better than their counterparts. These results suggest that focusing on **A**, which reflect standard speech production, is crucial for the assessment as shown in [13] where language-specific embeddings acted as assessment criterion.

The results of experiments with varying combinations of phoneme-fused embeddings highlight the importance of using both of the acoustic embeddings. While a configuration of an additional MHA with **(P⊗C)** applied to the arithmetically calculated embeddings achieves a PCC comparable to the best performance, the application of **(P⊗A)** does not exhibit any improvement. This indicates that typical speech patterns of NH children, or developmental speech errors reflected in the embeddings, also play an important role.

6. Conclusion

An automatic assessment model is proposed, for the first time, for evaluating the speech production skills of children with CIs. The model utilizes two acoustic embeddings from Wav2Vec2.0 acoustic encoders, each trained on speech of adults and children with NH, respectively, surpassing models relying on single acoustic embeddings by simultaneously considering the typical speech production patterns of both adults and NH children. Intensive experiments involving various combinations of the embeddings also underscore the importance of each acoustic embeddings and the significance of employing both embeddings.

Although the size of the dataset is relatively large for disordered speech research, the absolute size remains small, leading to a speaker-dependent approach. Therefore, future work should include experiments with additional data once the ratings by the experts are complete. In addition, it is necessary to explore which segmental and suprasegmental features influence the predicted scores and to what extent. Identifying erroneous fragments could also provide insights into how to improve children with CI's speech production skills.

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